Motivated by this viewpoint, we can see that dynamic version of Newton's formula is

$$f(x) = \sum_{k=0}^{\infty} \frac{(x \mid \alpha, \beta)_k}{k!} \cdot D_g f(a), \tag{5}$$

where a is an arbitrary integer, $D_g f(a)$ is dynamic difference operator, and $(x \mid \alpha, \beta)_n$ is generalized factorial which uses the combination of Nörlund's notation [19], and definition given by Steffensen [9]

Definition 11.1 (Generalized factorial)**.** *For non-negative integers x, n, α, β*

$$(x \mid \alpha, \beta)_n = x(x - n\alpha - \beta)(x - n\alpha - 2\beta) \cdots (x - n\alpha - n - I\beta).$$

For instance,

$$\begin{cases} (x)_n & = (x \mid \alpha, \beta)_n \,, \quad \alpha = 0, \quad \beta = 1, \quad \text{(forward Newton's formula)} \\ x^{(n)} & = (x \mid \alpha, \beta)_n \,, \quad \alpha = 0, \quad \beta = -1, \quad \text{(backward Newton's formula)} \\ x^{[n]} & = (x \mid \alpha, \beta)_n \,, \quad \alpha = \frac{1}{2}, \quad \beta = 1, \quad \text{(central Newton's formula)} \\ x^{n} & = (x \mid \alpha, \beta)_n \,, \quad \alpha = 0, \quad \beta = 0, \quad \text{(Taylor's formula)} \end{cases}$$

More precisely, the algorithm for closed formulas for sums of powers is